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SUPERSEDING
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MILITARY SPECIFICATION

NICKEL-COPPER ALLOY, WROUGHT; (55-60 PERCENT NICKEL) LOW PERMEABILITY

1. SCOPE

1.1 Scope. - This specification covers nickel-copper alloy used for applications requiring comparatively high strength and resistance to corrosion where a material with low magnetic permeability is required (see 6.3.6). The permeability of this alloy is, for all practical purposes, not affected by the normal fabricating and processing operations to which material may be subjected.

1.2 Classification. - Nickel-copper alloy shall be of the following forms and conditions as specified (see 6.2):

Bars and rods:
  Cold drawn.
  Hot finished.
Forgings: Hot finished.
Shapes: Hot finished.
Sheets:
  Cold rolled.
  Hot rolled.
Strips: Cold rolled.
Plates: Hot rolled.
Wire: Cold drawn.

1.3 Where no condition is specified, annealed material shall be furnished.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY
MIL-P-3993 - Preparation for Shipment and Storage of Copper and Copper Alloy Mill Products.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

FEDERAL
Fed. Std. No. 151 - Metals; Test Methods.

FSC 9525, 9530, 9535, 9540

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MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. -

3.1.1 The material used shall be such as to produce nickel-copper alloy in full compliance with the chemical, mechanical, and physical (magnetic permeability) requirements of this specification.

3.1.2 The material shall be uniform in quality and condition, clean, sound, smooth, commercially straight or flat, and free from foreign material, pipes, slivers, laps, cracks, twists, seams, burrs, damaged ends or edges, buckles, and other injurious defects.

3.2 Chemical requirements. - The material shall conform to the chemical requirements of Table I.

Table I - Chemical composition. [1]

| Nickel [2] | Iron (max.) | Manganese | Carbon (max.) | Silicon (max.) | Sulfur (max.) | Copper |
|---|---|---|---|---|---|---|
| Percent | Percent | Percent | Percent | Percent | Percent | Percent |
| 55 - 60 | 1.00 | 1.25 - 2.25 | 0.30 | 0.75 | 0.024 | Remainder |

[1] Where material is intended for sulfuric acid pickling service, iron shall be 2.50 percent max. and manganese 2.25 percent max.
[2] Cobalt counting as nickel.



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3.3 Mechanical properties. - Mechanical properties shall be as shown in tables II, III, and IV.

Table II - Mechanical properties - rods, bars and wire.

| Condition (temper) and diameter or distance between parallel faces (inch) | Tensile strength p.s.i. (min.) | Yield strength at 0.2 percent offset or at extension indicated | | Elongation in 2 inches (percent) (min.) |
|---|---|---|---|---|
| | | P.s.i. (min.) | Inch in 2 inches | |
| Rod and bar | | | | |
| Cold drawn, stress relieved: | | | | |
| Rounds, under 1/2 | 80,000 | 55,000 | 0.0084 | [1]10.0 |
| Rounds, 1/2 to 3-1/2, incl. | 80,000 | 55,000 | .0084 | 22.0 |
| Rounds, over 3-1/2 to 4, incl. | 75,000 | 45,000 | .0076 | 25.0 |
| Squares, hexagons and rectangles, all sizes | 75,000 | 45,000 | .0076 | [1,4]22.0 |
| Cold-drawn, annealed: | | | | |
| Rod and bar, all sizes | 65,000 | 23,000 | .0058 | [1]35.0 |
| Hot-finished: | | | | |
| Rounds, squares, rectangles, all sizes; also hexagons 2-1/8 and under | 75,000 | 35,000 | .0068 | 30.0 |
| Hexagons over 2-1/8 and shapes | 70,000 | 25,000 | .0060 | 30.0 |
| Rectangles, 5/16 thick and under | 75,000 | 35,000 | .0068 | 20.0 |
| Hot-finished, annealed; all sizes | 65,000 | 23,000 | .0058 | 35.0 |
| Forging quality[2] (all sizes) | ------ | ------ | ------ | ---- |
| Wire (in coil) | | | | |
| Cold-drawn:[3] | | | | |
| No. 0 temper, under 1/2 | 75/90,000 | ------ | ------ | ---- |
| No. 1 temper, under 1/2 | 85/100,000 | ------ | ------ | ---- |
| No. 2 temper, under 1/2 | 100/130,000 | ------ | ------ | ---- |
| No. 2 temper, 1/2 and over | 85/125,000 | ------ | ------ | ---- |
| Cold-drawn, annealed, all sizes | 65/85,000 | ------ | ------ | ---- |

[1] Not applicable for diameters or cross-sections under 0.093 inch.
[2] Furnished to chemical composition and surface inspection only. No tensile properties are required.
[3] Wire in straightened and cut lengths may have values slightly less than the minimum shown for No. 0, 1 and 2 tempers.
[4] Sizes under 1/2 inch may be 10.0 percent minimum.



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Table III - Mechanical properties of plate, sheet and strip.

| Condition | Tensile strength p.s.i. (min.) | Yield strength 0.2 percent offset or at extension indicated | | Elongation in 2 inches (percent) (min.) | Rockwell hardness[6] (B scale) |
|---|---|---|---|---|---|
| | | P.s.i., (min.) | Inch in 2 inches | | |
| **Hot rolled plate** | | | | | |
| Annealed | 65,000 | 23,000 | 0.0058 | 35 | [3]100 - 150 |
| As-rolled | 70,000 | 35,000 | .0068 | [7]30 | [3]130 - 170 |
| **Hot rolled, annealed and pickled sheet** | | | | | |
| Annealed | 65,000 | [1]23,000 | [3]0.0058 | 35 | 73 max. |
| **Cold rolled sheet** | | | | | |
| Annealed[4] | 65,000 | [1]23,000 | [3]0.0058 | 35 | 73 max. |
| 1/2 hard | ------ | ---- | ------ | ---- | 82 - 90 |
| **Cold rolled strip[5]** | | | | | |
| Annealed[4] | [2]65,000 to 85,000 | [1]23,000 | [3]0.0058 | [2]35 | 68 max. |
| Skin hard | ------ | ---- | ------ | ---- | 68 - 73 |
| 1/4 hard | ------ | ---- | ------ | ---- | 73 - 83 |
| 1/2 hard | ------ | ---- | ------ | ---- | 82 - 90 |
| 3/4 hard | ------ | ---- | ------ | ---- | 89 - 94 |
| Hard | ------ | ---- | ------ | ---- | 93 - 98 |
| Spring temper | ------ | ---- | ------ | ---- | 98 min. |

[1]Yield strength not required for sheet and strip under 0.020 inch thick.
[2]Not applicable for thicknesses under 0.010 inch.
[3]Numbers shown are Brinell hardness values; for information only.
[4]For deep drawing and spinning quality requirements see table IV.
[5]Accurate hardness determinations are difficult to make on strip under 0.005 inch in thickness and are not recommended.
[6]Hardness values are for information only where tensile requirements are shown; otherwise hardness is mandatory.
[7]As-rolled plate 5/16 inch thick and under, elongation 20 percent minimum.



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Table IV - Grain size[5] and hardness requirements for cold rolled, deep drawing and spinning quality sheet and strip.

| Thickness, inch | Calculated diameter of average grain section, max. | | Corresponding ASTM Micro grain size No. | Rockwell[3,4] hardness (B scale) (max.) |
|---|---|---|---|---|
| | Mm. | Inch | | |
| **Sheet (56 inches wide and under)** | | | | |
| 0.050 and less | 0.075 | 0.0030 | 4.5 | 68 |
| Over 0.050 to 0.250, incl. | .110 | .0043 | 3.5 | 68 |
| **Strip[1] (all widths)** | | | | |
| 0.005 to 0.015, incl. | 0.022 | 0.0009 | [2]8 | [2]73 |
| Over 0.015 to 0.024, incl. | .060 | .0024 | 5.5 | 70 |
| Over 0.024 to 0.125, incl. | .075 | .0030 | 4.5 | 68 |

[1] Grain size and hardness requirements other than those indicated above shall be agreed upon between the manufacturer and the purchaser.
[2] Accurate grain size and hardness determinations are difficult to make on strip under 0.005 inch in thickness and are not recommended.
[3] For material 0.050 inch and over in thickness, the Rockwell B hardness may be used. For material under 0.050 inch thick, the Rockwell superficial or Vickers hardness test is suggested.
[4] Rockwell or equivalent hardness; for hardness conversions, see Standard Fed. Std. No. 151.
[5] Determined in accordance with method No. 312 of Standard Fed. Std. No. 151.

3.4 Magnetic permeability. - The material specified herein shall have a permeability of 1.1 maximum at 27° Fahrenheit (F.) maximum and field strength of 0.5 oersted.

3.4.1 Permeability requirements shall apply to all material ordered unless otherwise agreed upon between the Government and the contractor (see 4.8 and 6.3.5).

3.5 Forgings. -

3.5.1 Dimensions. - All forgings other than forged rods and bars shall conform to the sizes and shapes as specified (see 6.2). The responsibility of furnishing forgings that can be laid out and machined to the finished dimensions within the specified tolerances as shown on the drawings and that will conform to such gages as may be specified in individual cases shall rest with the contractor. Sufficient stock shall be allowed for finishing. Forgings unreasonably in excess of the specified size or weight shall be rejected.



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3.5.2 Permissible variations. - When dimensional tolerances are not included in the contract or order, forgings measured on their diameters or between parallel faces shall not vary from the specified dimensions by more than the amounts shown in table V for rods and bars, and by not more than plus or minus 3/32 inch on smooth forged or plus or minus 1/32 inch on rough machined forgings other than rods and bars.

Table V - Permissible variations in diameter or distance between parallel surfaces of hot finished rods and bars.

| Specified dimensions, inches[1] | Permissible variations from specified dimensions, inch | |
|---|---|---|
| | Plus | Minus |
| Rods and bars, hot finished: | | |
| 1 and under | 0.016 | 0.016 |
| Over 1 to 2, incl. | .031 | .016 |
| Over 2 to 4, incl. | .047 | .031 |
| Over 4 | .125 | .063 |
| Round rods, hot finished, semi-smooth machined, over 4 inches | .031 | 0 |
| Round rods, hot finished, rough-turned, 1 inch and over | .031 | 0 |
| Round rods, hot finished, smooth finish machined over 4 inches | 0 | [2].005 |
| Forging quality rounds:[3] | | |
| Under 1 | .005 | .005 |
| 1 and over | .031 | 0 |
| Forging quality bolt stock (rounds only): | | |
| 1/4, 5/16 | 0 | .0062 |
| 3/8, 7/16, 1/2 | 0 | .0066 |
| 9/16 to 7/8, in 1/16 | 0 | .0082 |
| 15/16 and 1 | 0 | .0098 |
| 1-1/16 to 1-1/2 incl, in 1/16 | 0 | .0112 |

[1]Dimensions apply to diameter of rounds, to distance across flats of hexagons and squares, and separately to width and thickness of rectangles.

[2]Permissible variations available as plus 0.005 inch minus 0; or plus 0.0025 inch minus 0.0025 inch when so specified.

[3]Spot grinding may be permitted to remove minor surface defects. The depth of these ground areas shall not exceed 3 percent of the diameter of the rod.

3.6 Permissible variations, bars and rods. -

3.6.1 Cold drawn. -

3.6.1.1 Diameter, thickness, or width. - Cold drawn bars and rods, measured on their diameters or between parallel faces, shall not vary at any point from the specified dimensions by more than the amounts shown in table VI.



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3.6.1.2 Round rods shall not be out-of-round by more than one-half the permissible variation in diameter shown in table VI.

3.6.1.3 Square, rectangular, and hexagonal bars and rods shall have practically exact angles and shall have sharp corners unless otherwise specified.

Table VI - Permissible variations in diameter or distance between parallel surfaces of cold drawn bars and rods.

| Dimensions, inches[1] | Permissible variation, inch | |
|---|---|---|
| | Plus | Minus |
| Hexagons, squares, rectangles: | | |
| 1/16 to 3/16, excl. | 0 | 0.002 |
| 3/16 to 15/16, incl. | 0 | .003 |
| Over 15/16 to 1-1/2, incl. | 0 | .004 |
| Over 1-1/2 to 2, incl. | 0 | .005 |
| Over 2 to 2-3/4, incl. | 0 | .006 |
| Over 2-3/4 to 4, incl. | 0 | .007 |
| Rounds: | | |
| 1/16 to 3/16, excl. | 0 | .002 |
| 3/16 to 1/2, excl. | 0 | .003 |
| 1/2 to 15/16, incl. | 0 | .002 |
| Over 15/16 to 1-15/16, incl. | 0 | .003 |
| Over 1-15/16 to 2-1/2, incl. | 0 | .004 |
| Over 2-1/2 to 3, incl. | 0 | .005 |
| Over 3 to 3-1/2, incl. | 0 | .006 |
| Over 3-1/2 to 4, incl. | 0 | .007 |

[1]Dimensions apply to the diameter of rounds, distance across flats for hexagons and squares, and separately to width and thickness of rectangles.

3.6.1.4 Straightness. -

3.6.1.4.1 The permissible variations in straightness of cold drawn rods and bars as determined by the departure from true straightness (throw in one revolution or depth of chord) shall be as specified in table VII.

Table VII - Permissible variations in straightness of cold drawn rods and bars (see 6.3.4).

| Specified diameter or distance between parallel surfaces, inches | Permissible variations in lengths indicated, inch |
|---|---|
| Rounds: | Throw in one revolution: |
| 15/16 and under | 0.020 in 42 inch length |
| Over 15/16 to 4, incl. | 0.015 in 42 inch length |
| Squares, hexagons, flats: | Depth of chord: |
| All sizes | 0.025 per foot of length |



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3.6.1.4.2 The permissible variations in straightness of precision straightened cold drawn round rod and shafting as determined by the departure from straightness (throw in one revolution), shall be as specified in table VIII. All precision straightened rods shall be checked for straightness when supported on rollers at 42-inch intervals, and also on rollers at the ends of the rods in diameters and lengths as specified in table VIII.

Table VIII - Permissible variations in straightness of precision-straightened cold-drawn shafting (see 6.3.4).

| Specified diameter of shafting, inches | Standard distance between supports inches | Permissible variations (throw in one revolution) from straightness, inch |
|---|---|---|
| 1/2 to 15/16, incl. | 42 | 0.005 |
| Over 15/16 to 1-15/16, incl. | 42 | .006 |
| Over 1-15/16 to 2-1/2, incl. | 42 | .007 |
| Over 2-1/2 to 4, incl. | 42 | .008 |
| 3/4 to 15/16, incl. | Specified lengths of 3 to 10 feet | 0.004 plus 0.0025 for each foot or fraction thereof in excess of 3 feet |
| Over 15/16 to 4, incl. | Specified lengths of 20 feet and less | 0.005 plus 0.0015 for each foot or fraction thereof in excess of 3 feet |

3.6.1.4.3 The permissible variations in straightness of hot finished rods and bars as determined by the departure from straightness (throw in one revolution or depth of chord) shall be as specified in table IX.

Table IX - Permissible variations in straightness of hot finished rods and bars.[1]

| Condition and finish | Permissible variations, inch per foot |
|---|---|
| Rods and bars, hot finished | [2]0.050 |
| Round rods, hot finished, ground or rough turned | [3] .050 |
| Round rods, hot finished, semi-smooth machined | [4,5] .0031 |
| Round rods, hot finished, smooth machined | [4] .0015 |

[1]Not applicable to forging quality rods.
[2]The maximum curvature (depth of chord) shall not exceed the values indicated multiplied by length in feet.
[3]The throw in one revolution in any 3 feet of length or fraction thereof.
[4]The throw in one revolution in any 20 feet maximum length shall not exceed the values indicated.
[5]For shafts over 20 to 30 feet, the maximum deviation from straightness (throw in one revolution) will not exceed 0.062 inch over entire length when supported at ends.



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3.6.2 Hot finished bars and rods. -

3.6.2.1 Diameter, thickness, or width. - Hot rolled and forged bars and rods, measured on their diameter or between parallel surfaces, shall not vary at any point from the specified dimensions by more than the amounts shown in table V.

3.6.2.2 Spot grinding of forging stock, to remove minor defects, may be employed as shown in table V, note 3.

3.6.2.3 Round rods in straight lengths shall not be out-of-round by more than one-half the total permissible variation in diameter shown in table V except for rounds 1/2 inch and under, for which total permissible variation in diameter is permitted.

3.6.2.4 Square, rectangular, and hexagonal bars and rods shall have practically exact angles and shall have sharp corners consistent with good commercial practice. Where more precise angles and sharper corners are required, cold drawn rods and bars should be used, where available.

3.6.3 Length - bars and rods. -

3.6.3.1 Bars and rods of all sizes may be ordered in cut lengths, random lengths, nominal (stock) lengths, or in multiple lengths as specified (see 6.2).

3.6.3.2 Unless otherwise specified in the contract or order, bars and rods shall be furnished in random mill lengths with cut or sheared ends. Rods and bars furnished to nominal (stock) lengths shall be furnished with either cropped or saw-cut ends. Material ordered to cut lengths shall be furnished with square saw-cut or machined ends. Where rods and bars are ordered in multiple lengths, 1/4 inch will be allowed for each multiple cut unless otherwise specified in the contract or order, and the rods and bars will have cut ends.

3.6.3.3 The permissible variations in lengths of rods and bars, all conditions, shall be as specified in table X.

Table X - Permissible variations in length of bars and rods, all conditions.

| Specified length, feet, | Shortest acceptable length, feet | Maximum acceptable length | Maximum permissible percentage of weight of short lengths |
|---|---|---|---|
| Random mill lengths[1] | 6 | 24 feet | 25 percent under 9 feet |
| Nominal (stock) length[2] | | | |
| 16 to 18 | 16 | 18 feet, 1/2 inch | 100 percent 16 feet or longer |
| 14 to 16 | 14 | 16 feet, 1/2 inch | 100 percent 14 feet or longer |
| 12 to 14 | 12 | 14 feet, 1/2 inch | 100 percent 12 feet or longer |
| [3]10 to 12 | 10 | 12 feet, 1/2 inch | 100 percent 10 feet or longer |
| [3] 8 to 10 | 8 | 10 feet, 1/2 inch | 100 percent 8 feet or longer |
| [3] 6 to 8 | 6 | 8 feet, 1/2 inch | 100 percent 6 feet or longer |
| 4 to 6 | 4 | 6 feet, 1/2 inch | 100 percent 4 feet or longer |
| Cut-to-length | Specified length | Specified length to plus 1/8 inch | 100 percent specified length to plus 1/8 inch |

[1]Not applicable to "smooth-forged" rods and bars or to hot finished sections weighing over 25 pounds per lineal foot. These items are generally not available in the range of lengths indicated; short lengths down to 2 feet in lieu of 6 feet may be furnished.

[2]Other nominal lengths with a specified range of not less than 2 feet, with no shorts allowed, may also be furnished.

[3]For cold drawn rods and bars under 1/2 inch in diameter or distance across flats ordered to nominal or stock lengths with a 2 foot range, at least 93 percent of such material shall be within the range specified, the balance may be shorter lengths but in no case shall lengths less than 4 feet be furnished.



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3.7 Permissible variations, shapes. -

3.7.1 The permissible variations in cross-sectional dimensions of shapes shall be as specified (see 6.2).

3.7.2 Unless otherwise specified in the contract or order, shapes shall have practically exact angles and shall have sharp corners.

3.7.3 Length. - A variation of plus 1/8 inch will be permitted on shapes ordered to cut-to-length.

3.8 Permissible variations - plates, sheets, and strips. -

3.8.1 Thickness. - The permissible variations in thickness shall be as shown in tables XI, XII, XIII, and XIV.

Table XI - Permissible variations in thickness of sheets, all dimensions in inches.

| Specified thickness | Permissible variations[1] in thickness for widths given | | | |
|---|---|---|---|---|
| Sheet | | | | |
| | Hot-rolled | | Cold-rolled | |
| | 48 and under | Over 48 to 60, incl. | 48 and under | Over 48 to 60, incl. |
| | Plus and minus | Plus and minus | Plus and minus | Plus and minus |
| 0.018 to 0.025, incl. | 0.003 | 0.004 | 0.002 | 0.003 |
| Over 0.025 to 0.034, incl. | .004 | .005 | .003 | .004 |
| Over 0.034 to 0.043, incl. | .005 | .006 | .004 | .005 |
| Over 0.043 to 0.056, incl. | .005 | .006 | .004 | .005 |
| Over 0.056 to 0.070, incl. | .006 | .007 | .005 | .006 |
| Over 0.070 to 0.078, incl. | .007 | .008 | .006 | .007 |
| Over 0.078 to 0.093, incl. | .008 | .009 | .007 | .008 |
| Over 0.093 to 0.109, incl. | .009 | .010 | .007 | .009 |
| Over 0.109 to 0.125, incl. | .010 | .012 | .008 | .010 |
| Over 0.125 to 0.140, incl. | .012 | .014 | .008 | .010 |
| Over 0.140 to 0.171, incl. | .014 | .016 | .009 | .012 |
| Over 0.171 to 0.187, incl. | .015 | .017 | .010 | .013 |
| Over 0.187 to 0.218, incl. | .017 | .019 | .011 | .015 |
| Over 0.218 to 0.234, incl. | .018 | .020 | .012 | .016 |
| Over 0.234 to 0.250, incl. | .020 | .022 | .013 | .018 |

[1]Measured 3/8 inch or more from any edge.



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[illegible] may be detached, folded along the lines indicated, taped along the loose edge (DO NOT STAPLE)

Table XII - Permissible variations in thickness of cold rolled strips.

| Specified thickness, incl. | Permissible variations[1] inch for all widths |
|---|---|
| | Plus and minus |
| Under 0.006 | [2]10 |
| 0.006 to 0.015, incl. | 0.001 |
| Over 0.015 to 0.050, incl. | .0015 |
| Over 0.050 to 0.093, incl. | .0025 |
| Over 0.093 to 0.125, incl. | .004 |

[1] Measured 3/8 inch or more from any edge except for strip under 1 inch in width which is measured at any place.

[2] Expressed as percent of nominal thickness.



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Table XIII - Permissible variations in thickness and overweight of hot rolled rectangular plates up to 2 inches thick, inclusive.

Note. - All plates shall be ordered to thickness and not to weight per square foot. No plates shall vary more than 0.01 inch under the thickness ordered, and the overweight of each lot[1] in each shipment shall not exceed the amount shown in the table. Spot grinding will be permitted to remove surface imperfections, such spots not to exceed 0.01 inch under the specified thickness. This material shall be assumed to weigh 0.320 lb. per cu. inch.

| Specified thickness, inches | Permissible excess in average weight[2,3] per square foot of plates for widths given in inches expressed in percentage of nominal weights | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Under 48 | 48 to 60, excl. | 60 to 72, excl. | 72 to 84, excl. | 84 to 96, excl. | 96 to 108, excl. | 108 to 120, excl. | 120 to 132, excl. | 132 to 144, excl. | 144 to 160, incl. |
| 3/16 to 5/16, excl. | 9.0 | 10.5 | 12.0 | 13.5 | 15.0 | 16.5 | 18.0 | ---- | ---- | ---- |
| 5/16 to 3/8, excl. | 7.5 | 9.0 | 10.5 | 12.0 | 13.5 | 15.0 | 16.5 | 18.0 | ---- | ---- |
| 3/8 to 7/16, excl. | 7.0 | 7.5 | 9.0 | 10.5 | 12.0 | 13.5 | 15.0 | 16.5 | 18.0 | 19.5 |
| 7/16 to 1/2, excl. | 6.0 | 7.0 | 7.5 | 9.0 | 10.5 | 12.0 | 13.5 | 15.0 | 16.5 | 18.0 |
| 1/2 to 5/8, excl. | 5.0 | 6.0 | 7.0 | 7.5 | 9.0 | 10.5 | 12.0 | 13.5 | 15.0 | 16.5 |
| 5/8 to 3/4, excl. | 4.5 | 5.5 | 6.0 | 7.0 | 7.5 | 9.0 | 10.5 | 12.0 | 13.5 | 15.0 |
| 3/4 to 1, excl. | 4.0 | 4.5 | 5.5 | 6.0 | 7.0 | 7.5 | 9.0 | 10.5 | 12.0 | 13.5 |
| 1 to 2, incl. | 4.0 | 4.0 | 4.5 | 5.5 | 6.0 | 7.0 | 7.5 | 9.0 | 10.5 | 12.0 |

[1] The term "lot" applied to the above table means all of the plates of each "group width" and each "group thickness".
[2] Permissible overweight for lots of circular and sketch plates shall be 25 percent greater than the amounts given in the above table.
[3] The weight of individual plates shall not exceed the nominal weight by more than 1-1/4 times the amount shown in the table and in footnote 2.

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Table XIV - Permissible variations in thickness for rectangular plates over 2 inches in thickness.

| Specified thickness, inch | Variation over specified thickness for widths, given, inch | | | | | |
|---|---|---|---|---|---|---|
| | To 36 excl. | 36 to 60, excl. | 60 to 84, excl. | 84 to 120, excl. | 120 to 132, excl. | 132 and over |
| Over 2 to 3, excl. | 1/16 | 3/32 | 7/64 | 1/8 | 1/8 | 9/64 |
| 3 to 4, incl. | 5/64 | 3/32 | 7/64 | 1/8 | 1/8 | 9/64 |

3.8.1.1 The permissible thickness variations on sheets and strips shall be based on measurements taken with the micrometer spindle 3/8 inch or more from any edge for material 1 inch or over in width and at any place on strip under 1 inch in width and shall be as shown in tables XI and XII.

3.8.1.2 The permissible variations under the specified thickness and the permissible excess in overweight of plates shall be as shown in tables XIII and XIV.

3.8.2 Width. -

3.8.2.1 The permissible variations in width for sheets and strips shall be as shown in table XV and those for plates shall be as shown in table XVI.

Table XV - Permissible variations in width of sheets and strips.[1]

| Specified thickness, inch | Width, inch | Permissible variations, inch | |
|---|---|---|---|
| | | Plus | Minus |
| **Sheets** | | | |
| Sheared: 0.018 to 0.250, incl. | All widths | [1]0.125 | 0 |
| **Strips** | | | |
| Slit edge: | | | |
| 0.009 and less | All widths | 0.005 | 0.005 |
| Over 0.009 to 0.024, incl. | All widths | .007 | .007 |
| Over 0.024 to 0.075, incl. | 14 and less | .007 | .007 |
| Over 0.075 to 0.100, incl. | 14 and less | .009 | .009 |
| Over 0.100 to 0.125, incl. | 14 and less | .012 | .012 |
| Round or square edge strip: | | | |
| 0.071 to 0.125, incl. | 3 and less | .005 | .005 |

[1]When so specified, sheets shall be furnished with a tolerance of plus 0 with the above tolerance on the minus side or with one-half the above tolerance on the plus side and one-half on the minus side.



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Table XV

[illegible] in width[1] of sheared and abrasive cut rectangular plate.[2]

| Specified [illegible] incl. | Permissible variations in width for widths given, inches | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | to 30, incl. | | Over 30 to 72, incl. | | Over 72 to 108, incl. | | Over 108 to 144, incl. | | Over 144 to 160, incl. | |
| | Minus | Plus | Minus | Plus | Minus | Plus | Minus | Plus | Minus | Plus |
| Sheared:[3] | | | | | | | | | | |
| 3/16 to 5/[illegible] | 1/8 | 1/4 | 1/8 | 3/8 | 1/8 | 1/2 | 1/8 | --- | --- | --- |
| 5/16 to 1/[illegible] | 1/8 | 3/8 | 1/8 | 3/8 | 1/8 | 1/2 | 1/8 | 5/8 | 1/8 | --- |
| 1/2 to 3/4, | 1/8 | 3/8 | 1/8 | 1/2 | 1/8 | 5/8 | 1/8 | 3/4 | 1/8 | --- |
| 3/4 to 1, [illegible] | 1/8 | 1/2 | 1/8 | 5/8 | 1/8 | 3/4 | 1/8 | 7/8 | 1/8 | --- |
| 1 to 1-1/4, | 1/8 | 5/8 | 1/8 | 3/4 | 1/8 | 7/8 | 1/8 | 1 | 1/8 | --- |
| Abrasive cut: | | | | | | | | | | |
| 3/16 to 1-[illegible] | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | --- |
| Over 1-1/8 | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | --- |

[1] Permissible [illegible] for powder or inert arc cut plate shall be as agreed upon between the manufacture[illegible] agency concerned.

[2] Permissible [illegible]ed, powder or inert arc cut circular plate shall be as agreed upon between the bureau or agency concerned.

[3] Minimum shea[illegible]es for 3/4 inch thick and under and 20 inches for thicknesses over 3/4 inch.

[4] The minimum [illegible]is 2 inches and increases to 4 inches for thicker plates.

[5] Tolerances a[illegible]ches maximum lengths. For lengths over 240 inches an additional 1/16 inch i[illegible]s and minus.

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3.8.2.2 Unless otherwise specified in the contract or order strips shall be furnished with slit edges.

3.8.3 Length. -

3.8.3.1 All sheets and strips in straight lengths ordered cut-to-length shall have a permissible variation of plus 1/8 inch over the specified length.

3.8.3.2 When strips up to 14 inches wide are ordered in nominal (stock) lengths, the pieces shall be cut to the nominal length plus 1/8 inch as a maximum. The shortest acceptable lengths, the maximum permissible percentage by weight of short lengths, and the required percentage, by weight, of nominal lengths in any one shipment shall be as shown in table XVII.

Table XVII - Permissible variations in length of strips 14 inches wide and less.

| Nominal (stock) length | Required percentage by weight, of nominal length | Maximum permissible percentage by weight, of short lengths | | | | |
|---|---|---|---|---|---|---|
| | | Over 8 to 10 feet, inclusive | Over 6 to 8 feet, inclusive | Over 4 to 6 feet, inclusive | 2 to 4 feet, inclusive | Under 2 feet |
| Feet | Percent | Percent | Percent | Percent | Percent | |
| 10 | 60 | 40 | 30 | 20 | 10 | 0 |
| 8 | 70 | ---- | 30 | 20 | 10 | 0 |
| 6 | 80 | ---- | ---- | 20 | 10 | 0 |



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3.8.3.3 The permissible variations in length for plates shall be as shown in table XVIII.

Table XVIII - Permissible variations in length of sheared and abrasive cut[2] rectangular plate.[3]

| Specified thickness, inches | Permissible variation in length for lengths given, incl.[1] | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 60 and under | | Over 60 to 96, incl. | | Over 96 to 120, incl. | | Over 120 to 240, incl. | | Over 240 to 360, incl. | | Over 360 to 450, incl. | | Over 450 to 540, incl. | | Over 540 | |
| | Plus | Minus | Plus | Minus | Plus | Minus | Plus | Minus | Plus | Minus | Plus | Minus | Plus | Minus | Plus | Minus |
| Sheared:[4] | | | | | | | | | | | | | | | | |
| 3/16 to 5/16, excl. | 3/16 | 1/8 | 1/4 | 1/8 | 3/8 | 1/8 | 1/2 | 1/8 | 5/8 | 1/8 | 3/4 | 1/8 | 7/8 | 1/8 | 7/8 | 1/8 |
| 5/16 to 1/2, excl. | 3/8 | 1/8 | 1/2 | 1/8 | 1/2 | 1/8 | 1/2 | 1/8 | 5/8 | 1/8 | 3/4 | 1/8 | 7/8 | 1/8 | 1 | 1/8 |
| 1/2 to 3/4, excl. | 1/2 | 1/8 | 1/2 | 1/8 | 5/8 | 1/8 | 5/8 | 1/8 | 3/4 | 1/8 | 7/8 | 1/8 | 1-1/8 | 1/8 | 1-3/8 | 1/8 |
| 3/4 to 1, excl. | 5/8 | 1/8 | 5/8 | 1/8 | 5/8 | 1/8 | 3/4 | 1/8 | 7/8 | 1/8 | 1-1/8 | 1/8 | 1-3/8 | 1/8 | 1-5/8 | 1/8 |
| 1 to 1-1/4, incl. | 3/4 | 1/8 | 3/4 | 1/8 | 3/4 | 1/8 | 7/8 | 1/8 | 1-1/8 | 1/8 | 1-3/8 | 1/8 | 1-5/8 | 1/8 | --- | --- |
| Abrasive cut:[2] | | | | | | | | | | | | | | | | |
| 5/16 to 1-1/2, incl. | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | 1/16 | --- | --- | --- | --- |
| Over 1-1/2 to 4, incl. | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | 1/8 | --- | --- | --- | --- |

[1] Permissible variations in length for powder or inert arc cut plate shall be as agreed upon between the manufacturer and the bureau or agency concerned.
[2] Abrasive cut applicable to maximum length of 144 or 400 inches depending on thickness and width ordered.
[3] Permissible variations in machined, powder or inert arc cut circular plate shall be as agreed upon between the manufacturer and the bureau or agency concerned.
[4] Minimum sheared length, 10 inches.

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3.8.3.3.1 Unless otherwise specified in the contract or order, plates 1-1/4 inches thick and under shall be furnished with a sheared edge, and plates over 1-1/4 inches shall be furnished with abrasive cut edges.

3.8.4 Diameter. -

3.8.4.1 The permissible variations in diameter of sheared circular plates shall be as shown in table XIX.

Table XIX - Permissible variations over specified diameter for thickness given.[1]

| Specified diameter inches | Thickness, inch to 3/8, incl. |
|---|---|
| 20 to 32, excl. | 1/4 |
| 32 to 84, excl. | 5/16 |
| 84 to 108, excl. | 3/8 |
| 108 to 130, excl. | 7/16 |

[1]No tolerance under.

3.8.5 Flatness. -

3.8.5.1 There shall be no flatness requirements for "deep drawing", "spinning quality" and "tempered" sheets and strips.



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3.8.5.2 The permissible variation in flatness for plates shall be as shown in table XX.

Table XX - Permissible variations in flatness of rectangular, circular and sketch plates.

| Specified thickness, incl. | Flatness tolerance (deviation from a horizontal flat surface), inch, for thicknesses and widths given[1, 2, 3, 4] | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | To 48, excl. | 48 to 60, excl. | 60 to 72, excl. | 72 to 84, excl. | 84 to 96, excl. | 96 to 108, excl. | 107 to 120, excl. | 120 to 144, excl. | 144 and over |
| 3/16 to 1/4, excl. | 3/4 | 1-1/16 | 1- 1/4 | 1-3/8 | 1- 5/8 | 1- 5/8 | ----- | ---- | ----- |
| 1/4 to 3/8, excl. | 11/16 | 3/4 | 15/16 | 1-1/8 | 1- 3/8 | 1- 7/16 | 1- 9/16 | 1-7/8 | ----- |
| 3/8 to 1/2, excl. | 1/2 | 9/16 | 11/16 | 3/4 | 15/16 | 1- 1/8 | 1- 1/4 | 1-7/16 | 1-3/4 |
| 1/2 to 3/4, excl. | 1/2 | 9/16 | 5/8 | 5/8 | 13/16 | 1- 1/8 | 1- 1/8 | 1-1/8 | 1-3/8 |
| 3/4 to 1, excl. | 1/2 | 9/16 | 5/8 | 5/8 | 3/4 | 13/16 | 15/16 | 1 | 1-1/8 |
| 1 to 2, excl. | 1/2 | 9/16 | 9/16 | 9/16 | 11/16 | 11/16 | 11/16 | 3/4 | 1 |
| 2 to 4, incl. | 1/4 | 5/16 | 3/8 | 7/16 | 1/2 | 9/16 | 5/8 | 3/4 | 7/8 |

[1] Tolerances apply to plates up to 12 feet length or to any 12 feet of longer plates.
[2] If the longer dimension is under 36 inches, the tolerance is not greater than 1/4 inch.
[3] The shorter dimension specified is considered the width and the flatness deviation across the width does not exceed the tabular amount for that dimension.
[4] The maximum deviation from a horizontal flat surface does not customarily exceed the tabular tolerance for the longer dimension specified.

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3.8.6 Curvature. - Sheets and strips in straightened and cut lengths shall not exhibit edgewise curvature (depth of chord) in excess of 0.05 inch per foot of length.

3.8.7 Edges (strip). -

3.8.7.1 When no description of any required form of edge on strip is given, it shall be understood that edges such as would result from slitting will be required.

3.8.8 Unless otherwise specified in the contract or order, sheets shall be furnished to the specified widths and lengths; strip shall be furnished in random (straight) lengths; plates 1-1/4 inches in thickness and less shall be furnished with sheared edges; and plates over 1-1/4 inches in thickness shall be furnished with abrasive cut edges.

3.9 Permissible variations, wire. -

3.9.1 Diameter. - The permissible variations in diameter and out-of-roundness of cold drawn wire shall be as shown in table XXI.

Table XXI - Permissible variations in diameter of cold drawn wires.

| Diameter inch | Permissible variations[1] inch | |
|---|---|---|
| | Plus | Minus |
| 0.002 to 0.0044, incl. | 0.0002 | 0.0002 |
| Over 0.0044 to 0.0079, incl. | .00025 | .00025 |
| Over 0.0079 to 0.0149, incl. | .0003 | .0003 |
| Over 0.0149 to 0.0199, incl. | .0004 | .0004 |
| Over 0.0199 to 0.031, incl. | .0005 | .0005 |
| Over 0.031 to 0.045, incl. | .0006 | .0006 |
| Over 0.045 to 0.079, incl. | .0007 | .0007 |
| Over 0.079 to 0.1875, incl. | .001 | .001 |
| Over 0.1875 to 0.406, incl. | .0015 | .0015 |
| Over 0.406 | .002 | .002 |

[1]Wires shall not be out-of-round by more than one-half the total permissible variations shown.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 The requirements of 4.2.1 through 4.10.1 inclusive shall be applied when the Government inspector determines that a manufacturer's method of quality control is not acceptable to the Government.

4.2.1 Sampling, examination, and testing shall be in accordance with Standard Fed. Std. No. 151 and as specified herein.



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4.3 Lot. -

4.3.1 For visual examination and mechanical tests. - Unless otherwise specified in the contract or order, a lot shall consist of material of one condition, and size offered for delivery at one time. (For mechanical tests, size shall refer to the same thickness of sheet, strip, or plate and the same cross section of other products.) In addition, when inspection is made at the place of manufacture, a lot shall consist of material from only one heat.

4.3.2 For chemical analysis. - A lot shall consist of material from one heat. Where material cannot be identified by heat, a lot shall consist of not more than 500 pounds of material offered for delivery at one time.

4.3.3 For magnetic permeability test. - Unless otherwise specified in the contract or order, a lot shall consist of material from one heat.

4.4 Sampling for chemical analysis. - One representative sample shall be obtained from each lot that can be identified by heat. Where material cannot be identified by heat, 2 samples shall be obtained. The samples shall be selected in a manner calculated to disclose any nonuniformity of material within the lot. The samples selected shall consist of not less than 2 ounces of chips or drillings. The samples shall be forwarded to a laboratory designated by the bureau or agency concerned.

4.5 Samples for check analysis of individual lots or shipments may be waived at the discretion of the procuring agency or the inspector provided the manufacturer's method of quality control is acceptable to the procuring agency or the inspector, as applicable, or that all of the material under inspection can be identified as being made from metal previously analyzed and found to be in conformance with the chemical composition specified in this specification.

4.6 Visual and dimensional examination. -

4.6.1 Forgings weighing 250 pounds or more. - Each forging weighing 250 pounds or more shall be subjected by the inspector to surface examination for workmanship and dimensions.

4.6.2 Straightness. - In determining straightness in the standard 42 inch distance between supports or in determining straightness when supported at the ends, the rod shall be placed on a precision table equipped with ball-bearing rollers and a micrometer or dial indicator. The rod shall then be rotated slowly against the indicator, and the deviation from straightness in any portion of the rod between the supports shall not exceed the permissible variations specified in tables VII and VIII. The deviation from straightness (throw in one revolution) is defined as the difference between the maximum and minimum readings of the dial indicator in one complete revolution of the rod.



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4.6.3 Small forgings and all other forms. -

4.6.3.1 Sampling. - From each lot, the inspector shall select representative samples for examination of visual and dimensional characteristics with lot acceptance based on the sampling inspection requirements shown in table XXII in accordance with Standard MIL-STD-105. In case of wire a unit shall be one coil of wire.

Table XXII - Sampling for visual and dimensional examination
AQL (approx.) = 1.5 percent defective.

| Number of units in lot | Number of units in sample | Acceptance number (defectives) | Rejection number (defectives) |
|---|---|---|---|
| 15 and under | 7 | 0 | 1 |
| 16 to 40 | 15 | 0 | 1 |
| 41 to 110 | 25 | 1 | 2 |
| 111 to 300 | 35 | 1 | 2 |
| 301 to 500 | 50 | 2 | 3 |
| 501 and over | 75 | 3 | 4 |

4.6.3.2 Each sample selected in accordance with 4.6.3.1 shall be subjected by the inspector to surface examination for workmanship and dimensions. Any sample unit containing one or more visual or dimensional defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. A rejected lot may be resubmitted for examination provided the manufacturer, after having been informed of the reasons for rejection, has examined each unit in the lot for the deficiency noted and has removed, reworked, or repaired all nonconforming units.

4.7 Sampling for mechanical tests. -

4.7.1 Tension tests. -

4.7.1.1 Where material can be identified by melt. - Unless otherwise specified in the contract or order, from each 5,000 pounds of material in the lot, the inspector shall take one tension test specimen.

4.7.1.2 Where material cannot be identified by melt. - From each 500 pounds of material in the lot, the inspector shall take one tension test specimen.

4.7.2 Bend test (wire only). - Where material can be identified by melt, the inspector shall take one best test specimen from each 5,000 pounds in the lot. Where material cannot be identified by melt, the inspector shall select one bend test specimen from each 500 pounds in the lot.

4.7.3 Rejection. - If any specimen fails to meet the requirements of this specification, the entire lot shall be rejected, subject to the retest provisions in accordance with Standard Fed. Std. No. 151.

4.8 Mechanical tests. -

4.8.1 Bars, forgings, rods, and shapes. - All bars, forgings, rods, and shapes shall be pulled in full size when practicable. When a machined specimen becomes necessary, enough metal may be removed from the gage section to meet the limitations of the testing machine or the specimen may be machined to the form and dimensions specified in Standard Fed. Std. No. 151.



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4.8.1.1 For bars, forgings, rods, and shapes up to 1-1/2 inches in diameter or thickness the axis of the test specimen shall coincide with the central axis of the piece; over 1-1/2 inches the axis shall be located midway between the center and surface of the piece.

4.8.1.2 The axis of tension test specimens for bars, forgings, rods, and shapes shall be parallel to the direction of rolling, drawing, or flow of metal.

4.8.1.3 Unless otherwise specified in the contract or order, rods and bars intended for reforging will not require physical tests.

4.8.2 Plates, sheets, and strips. -

4.8.2.1 Tension test specimens for plates, sheets, and strips not over 3/8 inch thick shall be machined to the form and dimensions of F2 in Standard Fed. Std. No. 151.

4.8.2.2 Tension test specimens for plates over 3/8 inch thick shall be machined to the form and dimensions of F1 in Standard Fed. Std. No. 151.

4.8.2.3 Tensile specimens from plates, sheets, and strips shall be taken transverse to the direction of rolling when width will permit.

4.8.3 Wire. -

4.8.3.1 Tension test specimens shall be of the full cross-section of the wire and not less than 15 inches in length. Specimens shall be free from sharp bends or kinks. The distance between the jaws of the testing machine, with the specimen in place ready for testing, shall be not less than 10 inches.

4.8.3.2 Bend test specimens shall be full size sections of wire of a suitable length. The specimens shall be wrapped at least 360 degrees around a rod of the same diameter as the wire and examined for cracks.

4.8.4 Yield strength. - The yield strength shall be determined by either the offset method or the extension under load method.

4.9 Permeability test. -

4.9.1 Sampling for magnetic permeability test. - From each lot specified in 4.3.3, the manufacturer shall select a representative pilot test sample and process to a hot finished bar. The bar shall then be heat treated to a condition which will simulate the maximum permeability to be expected from the mill processing of all forms, sizes and conditions (tempers) produced from this heat. The heat treatment to be used for this test specimen shall consist of heating the bar to approximately 1250°F. for 1 hour, followed by slow furnace cooling to approximately 800°F., then either air or furnace cooling. A sample machined from this bar shall then be tested under the conditions specified in 4.9.2.

NOTE: The heat treatment described above is intended only as a means for the manufacturer to determine the capability of a heat to meet the maximum permeability requirements in all the various forms, sizes and conditions (tempers) that might be produced from this heat and is not recommended heat treatment for production parts. In the event that the bureau or agency concerned elects to make his own check permeability determination on finished material, acceptance or rejection of this material shall be based on testing of this material in the condition (temper) in which it is purchased.

4.9.2 Permeability test. - Permeability tests shall be performed at a maximum temperature of 27°F. and a field strength of 0.5 oersteds direct current (d.c.).



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4.10 Test reports. - Test report shall consist of a statement to the effect that material complies with the requirements of the specification and the manufacturer shall also include in the test report the chemical (ladle) analyses of each heat and the results of mechanical property and permeability tests for each lot of materials furnished.

4.10.1 Certification of chemical composition and permeability by converters (rerollers and redrawers) shall be based on the results of tests reported by the manufacturer.

4.11 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - The nickel-copper alloy shall be packaged, packed and marked for the level as specified (see 6.2) in accordance with Specification MIL-P-3993.

6. NOTES

6.1 Intended use. -

6.1.1 Annealed, cold drawn rods are intended for severe cold working operations such as redrawing, swaging, and upsetting.

6.1.2 Annealed wire and intermediate temper wires other than "annealed" or "as drawn" are intended for the manufacture of cold headed bolts, rivets, screws, and nails and for roll threading and similar purposes. The selection of the proper temper will vary, depending on the type of heating machine used and the severity of cold forming involved, and the properties of such wires shall be as specified or as otherwise agreed upon between the wire-drawer and the fabricator.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Form, size, condition and surface finish of material required (see 1.2, tables II, III, and IV and 6.3.7).
(c) The application of the material; i.e. forging (see 3.5.1) forging quality rods for subsequent hot manipulation, cold heading (see 6.1.2).
(d) The lengths required and whether "cut-to-length", "random mill" lengths, "nominal (stock)" lengths, or "multiples" of specified lengths (see 3.6.3, 3.7.3, and 3.8.3).
(e) Dimensions of shapes (see 3.7.1).
(f) When tolerances other than those shown are required.
(g) The thickness of plates, sheets, and strips in decimals of an inch.
(h) Whether precision straightened, cold drawn shafting is required.
(i) Whether plate shall be furnished with sheared or abrasive cut edges.
(j) Whether permeability tests are required (see 4.3.3 and 6.3.5).
(k) Applicable levels of packaging and packing (see 5.1).

6.3 General information. -

6.3.1 When material is ordered in the form of plates, sheets, or strips, it shall be understood that these terms refer merely to the general form and dimensions of the material and do not have any technical significance as to the methods of manufacture.

6.3.2 Plates, sheets, and strips should be ordered in as narrow widths as can be used.

6.3.3 Hot rolled "forging quality" rounds (see tables II and V) with a cold drawn pass are available to special close tolerances for hot upsetting of bolts.



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6.3.4 The commercial straightness tolerances of cold drawn rods and bars are shown in table VII and those of precision straightened shafting are shown in table VIII. Where material is intended for shafting, the use of precision straightened cold drawn shafting is recommended (see 3.6.1.4).

6.3.5 Nickel-copper alloy (55-60 percent nickel) is similar to Specification QQ-N-281, except the the allow with the 55-60 nickel has low magnetic permeability and lower mechanical properties than nickel-copper alloy Specification QQ-N-281 and cognizance of these differences should be taken in design and in use where low permeability is desired. The modulus of elasticity for nickel-copper alloy (55-60 percent nickel) is 25,000,000 p.s.i. as compared to 26,000,000 p.s.i. for nickel-copper alloy Specification QQ-N-281. The corrosion resisting characteristics of the two materials are of the same order except for hot sulfuric acid pickling solutions. For the latter service the 55-60 percent Ni-Cu alloy is preferred to Specification QQ-N-281 as the latter material may, under certain environment, be susceptible to hydrogen embrittlement.

NOTE: Where material is intended for sulfuric acid service the contract or requisition should state "for pickling applications permeability test not required".

6.3.6 This nickel-copper alloy has found useful application for service applications requiring good strength, brazability, weldability and formability, resistance to corrosion, and low permeability, such as minesweeper fuel and oil tanks, nails and fastenings, cable sheathing, sulfuric acid pickling equipment and electronic applications, such as cathode gun structures and mass spectrometer parts.

6.3.7 All hot finished material, in the annealed condition, is available with a pickled or descaled finish. Hot rolled, annealed sheet is furnished with a pickled finish. Hot finished bars, rods, and shapes, in the annealed condition, may be furnished with a pickled or descaled finish. Hot rolled plate either in the as-rolled or annealed conditions, is available with a descaled (mineral shot blasted) surface. A descaled surface is generally preferred for welded parts, unless the edges to be welded are to be ground prior to welding, and particularly for applications where forming operations are involved and a good surface is required.



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6.4 Material is available commercially as shown in table XXIII and in the nominal ranges of sizes as shown in table XXIII. These are informative only and subject to change and the manufacturer should be consulted for latest information available.

Table XXIII - Conditions and finishes.

| Condition and form | Diameter or distance across flats, inches |
|---|---|
| **Rod (rounds)** | |
| Hot-finished: | |
| As hot-finished | 1/4 to 4-1/2, incl. |
| Semi-smooth machined | Over 4 to 10, incl. |
| Smooth-finished machined | Over 4 to 10, incl. |
| Rough-turned or ground[1] | 1/2 to 12, incl. |
| Bolt tolerance | 1/4 to 1-1/2, incl. |
| Cold-drawn | 1/16 to 4, incl. |
| **Bar** | |
| Hot-finished: | |
| Rectangles | 1/8 to 6, incl. by 1/2 to 10, incl. |
| Squares | 3/8 to 6, incl. |
| Hexagons | 3/8 to 8, incl. |
| Cold-drawn: | |
| Rectangles: | |
| Thicknesses | 1-3/4 and under |
| Widths | 4 and under |
| Squares | 2-1/8 and under |
| Hexagons | 2 and under |

[1]For forging quality material, spot grinding is permitted; however, when machining quality is specified, spot grinding is not permitted.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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